

Integer participation and structural noise in superconducting clusters

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We analyze a curated dataset of more than two hundred superconducting materials and liquid ⁴He using a single phenomenological observable, the “mother frequency” $F_m = \Theta_D/T_c$, where Θ_D is an effective Debye temperature and T_c the transition temperature. Using a structural-noise model inferred from previous work on cluster topology, frictional heating and pressure sensitivity, we define an ideal transition temperature $T_c^{\text{ideal}} = T_c(1 + \hat{\xi})$ and a noise-corrected frequency $F_m^* = \Theta_D/T_c^{\text{ideal}}$. We then search, for each family of materials, a base frequency f_{base} that minimizes the median distance of $N_{\text{corr}} = F_m^*/f_{\text{base}}$ to the nearest integer. The resulting integer “participation numbers” N_{corr} exhibit a strong deviation from several null models obtained by shuffling Θ_D or sampling from smooth continuous distributions. The distribution of $|\delta| = |N_{\text{corr}} - \text{round}(N_{\text{corr}})|$ is significantly more concentrated near zero than the nulls (Kolmogorov–Smirnov and Mann–Whitney $p \lesssim 10^{-13}$, Cliff’s $\Delta \simeq 0.3$), indicating genuine integer locking rather than a discretization artefact. Across families, we find a robust hierarchy of base frequencies: iron-based superconductors cluster near $f_{\text{base}} \simeq 1$, effectively behaving as natural, unscaled harmonic oscillators for which F_m^* itself is already an (almost) integer-valued participation number; binary, molecular and high-pressure materials sit near $f_{\text{base}} \simeq 4$, and oxides and heavy fermions around $f_{\text{base}} \simeq 2$. The global median $f_{\text{base}} \simeq 4.16$ is numerically close to the $n = 4$ thermal ratio previously identified between the superfluid and roton branches of liquid ⁴He. Finally, we show that materials with almost-integer participation systematically exhibit lower structural noise: the top 20% with smallest $|\delta|$ have a substantially lower noise z -score than the rest of the dataset (Cliff’s $\Delta \simeq -0.47$). Together, these results suggest that the mother frequency F_m probes a discrete oscillator structure underpinning both superconductivity and superfluidity, with family-dependent harmonics shaped by structural disorder and pressure.

I. INTRODUCTION

The search for organizing principles in the zoo of superconducting materials typically proceeds through microscopic models for pairing, band structure or electron–phonon coupling.[1, 2] Here we explore a complementary, purely phenomenological direction, inspired by methods in non-equilibrium statistical mechanics and dynamical systems [3–6]: we treat each superconductor or superfluid as a black box characterized by a single dimensionless frequency

$$F_m = \frac{\Theta_D}{T_c}, \quad (1)$$

constructed from an effective Debye temperature Θ_D and the observed critical temperature T_c . Previous work showed that, for liquid ⁴He, ratios of such frequencies across well-defined dynamical regimes (normal fluid, superfluid, roton) fall extremely close to small integers, with a particularly prominent factor $n = 4$ associated with the onset of the roton branch. This suggests that, at least for ⁴He, the spectrum of collective excitations organizes into discrete harmonic “shells”.

In this paper we ask whether an analogous discrete structure can be detected, in a data-driven way, across a heterogeneous ensemble of superconductors. Rather than committing to a specific microscopic Hamiltonian, we use F_m as a coarse-grained probe and search for hidden integer structure once known sources of noise are removed.

Two ingredients are essential: (i) a structural-noise model that accounts for deviations of the measured T_c

from an ideal value set by frictional losses, topology and pressure; and (ii) a robust calibration of a base frequency f_{base} that captures the dominant periodicity of F_m within each family. After correcting for noise, we show that the rescaled quantities $N_{\text{corr}} = F_m^*/f_{\text{base}}$ cluster close to integers far more strongly than expected from smooth null models. We further identify a family-dependent hierarchy of f_{base} that lines up convincingly with the integer ratios previously observed in ⁴He.

The rest of this paper is organized as follows. Sec. II describes the dataset and the structural-noise model. Sec. III defines the base-frequency calibration and the integer-participation metrics. Sec. IV presents the statistical evidence for integer locking and its dependence on structural noise and material family. Sec. V discusses physical interpretations and limitations, and Sec. VI summarizes the main conclusions. Additional figures validating the noise model are provided in the Appendix.

II. DATA AND STRUCTURAL-NOISE MODEL

A. Dataset and family labels

We work with the same curated dataset of superconducting materials used in previous studies of cluster topology and pressure sensitivity. Each entry corresponds to a material or fluid labeled by a name and a category: SC_TypeI, SC_TypeII, SC_Binary, SC_Oxide, SC_Molecular, SC_HighPressure, SC_HeavyFermion, SC_IronBased, and Superfluid (for liquid ⁴He). For

each entry i we have an effective Debye temperature $\Theta_D^{(i)}$, a measured transition temperature $T_c^{(i)}$, a family label $k(i)$, and a structural-noise prediction $\hat{\xi}^{(i)}$ obtained from a separate model that encodes frictional dissipation, topological defects and pressure-induced distortions.

B. Structural noise and ideal transition temperature

The structural-noise field $\hat{\xi}^{(i)}$ is defined such that $\hat{\xi}^{(i)} \geq 0$ corresponds to a loss of effective ordering relative to an otherwise similar but better structured material. We therefore interpret it as a multiplicative distortion of the transition temperature and define an “ideal” transition

$$T_c^{\text{ideal}(i)} = T_c^{(i)} [1 + \hat{\xi}^{(i)}]. \quad (2)$$

By construction, $T_c^{\text{ideal}(i)} \geq T_c^{(i)}$ and the correction vanishes for structurally optimal samples.

The corresponding noise-corrected mother frequency is

$$F_m^*(i) = \frac{\Theta_D^{(i)}}{T_c^{\text{ideal}(i)}}. \quad (3)$$

Fig. 8 (Appendix) confirms that the relation between T_c and T_c^{ideal} is close to the identity for most materials, with a small subset receiving sizable upward corrections.

To compare noise levels across families with different baselines, we standardize the predicted noise within each family k :

$$z_\xi^{(i)} = \frac{\hat{\xi}^{(i)} - \mu_{\xi,k(i)}}{\sigma_{\xi,k(i)}}, \quad (4)$$

where $\mu_{\xi,k}$ and $\sigma_{\xi,k}$ are the mean and standard deviation of $\hat{\xi}$ in family k . The resulting distributions of z_ξ are shown in Fig. 7 (Appendix).

III. BASE-FREQUENCY CALIBRATION AND INTEGER PARTICIPATION

A. Base frequency per family

We seek, for each family k , a base frequency $f_{\text{base}}^{(k)}$ such that the rescaled frequencies

$$N_{\text{corr}}^{(i)} = \frac{F_m^*(i)}{f_{\text{base}}^{(k(i))}} \quad (5)$$

cluster near integers. To quantify this we define the integer locking residual

$$\delta^{(i)} = N_{\text{corr}}^{(i)} - \text{round}(N_{\text{corr}}^{(i)}), \quad (6)$$

and focus on its absolute value $|\delta^{(i)}|$.

For each family k we choose $f_{\text{base}}^{(k)}$ by minimizing a robust loss

$$\mathcal{L}_k(f_{\text{base}}) = \text{median}_{i \in k} |N_{\text{corr}}^{(i)} - \text{round}(N_{\text{corr}}^{(i)})| + \lambda \text{median}_{i \in k} N_{\text{corr}}^{(i)}, \quad (7)$$

over a physically motivated range $0.5 \leq f_{\text{base}} \leq 5.0$. This explicitly rules out unphysical, high-frequency solutions (e.g., $f_{\text{base}} \sim 10^2\text{--}10^3$) that would artificially fit higher harmonics while providing no interpretable physical scale. The first term in Eq. (7) drives the residuals toward integers; the second is a small regularizer that penalizes large values of N_{corr} and selects low-order harmonics when multiple choices of f_{base} give similar residuals. The optimization is performed on a dense grid over the allowed range. We repeat the calibration under two hypotheses, using either F_m^* or $F_m^*/2$ as the target frequency, and keep the one with lower median residual. All results reported below use this per-family calibration.

B. Null models

To assess statistical significance, we compare the observed locking residuals $|\delta^{(i)}|$ with several null models that destroy any discrete structure while preserving broad distributional features of F_m^* [7].

The simplest null shuffles the pairing between Θ_D and T_c : we randomly permute Θ_D across the dataset, recompute F_m and F_m^* using Eqs. (2)–(3), recalibrate $f_{\text{base}}^{(k)}$ and recompute N_{corr} and $|\delta|$. This preserves the marginal distributions of Θ_D and T_c and the family labels, but destroys any material-wise correlation.

We also consider smooth parametric nulls in which F_m^* is sampled from a fitted gamma or log-normal distribution within each family, again followed by the same calibration procedure. In all cases we generate 10^3 realizations and pool the resulting $|\delta|$ values.

C. Effect-size and correlation metrics

We quantify deviations between the real and null distributions of $|\delta|$ using non-parametric tests: the Kolmogorov–Smirnov (KS) statistic, the Mann–Whitney (MW) rank-sum test, and Cliff’s Δ as a measure of effect size [8]. Cliff’s Δ is defined as

$$\Delta = \Pr(|\delta|_{\text{real}} < |\delta|_{\text{null}}) - \Pr(|\delta|_{\text{real}} > |\delta|_{\text{null}}), \quad (8)$$

and ranges from -1 to $+1$.

To connect locking strength with structural noise we compute, over all materials with $N_{\text{corr}}^{(i)} \geq 1$, the Spearman rank correlation between $|\delta^{(i)}|$ and $z_\xi^{(i)}$. We also split the dataset into the “almost-integer” group (top 20% lowest $|\delta|$) and the rest, and compare their z_ξ distributions using Cliff’s Δ .

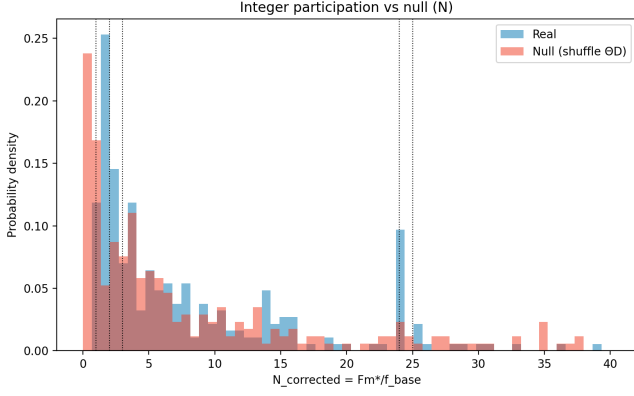


FIG. 1. Distribution of noise-corrected participation numbers $N_{\text{corr}} = F_m^*/f_{\text{base}}$ (“Real”) compared with a null model obtained by shuffling Θ_D across materials (“Null”). Vertical dotted lines highlight modes around $N \simeq 1$, $N \simeq 2$ and $N \simeq 24$ –25.

IV. RESULTS

A. Integer participation vs null models

Fig. 1 shows the empirical distribution of N_{corr} compared with its shuffled- Θ_D null. The real data exhibits clear accumulation near small integers $N \simeq 1$ and $N \simeq 2$ and a pronounced cluster near $N \simeq 24$ –25, which is largely absent in the null. These modes correspond to families whose calibrated f_{base} select low-order harmonics.

A more direct comparison is obtained by looking at the residuals $|\delta|$, shown in Fig. 2. The real distribution is significantly more concentrated near zero than the shuffled null: both KS and MW tests yield $p \lesssim 10^{-13}$, and Cliff’s $\Delta \simeq 0.32$ indicates a medium-to-large effect size. In other words, after correcting for structural noise, the dataset contains substantially more near-integer participation numbers than expected from a smooth frequency spectrum.

B. Family-dependent locking strength

The locking residuals vary systematically across families. Fig. 3 summarizes the distribution of $|\delta|$ (clipped at 1.0 for readability) for each category. Iron-based superconductors and heavy-fermion systems show the smallest median residuals, indicating strong integer participation, whereas type-I and high-pressure materials exhibit broader tails.

C. Structural noise vs integer locking

We now relate locking strength to structural noise. Fig. 4 plots $|\delta^{(i)}|$ against the noise z -score $z_\xi^{(i)}$, with

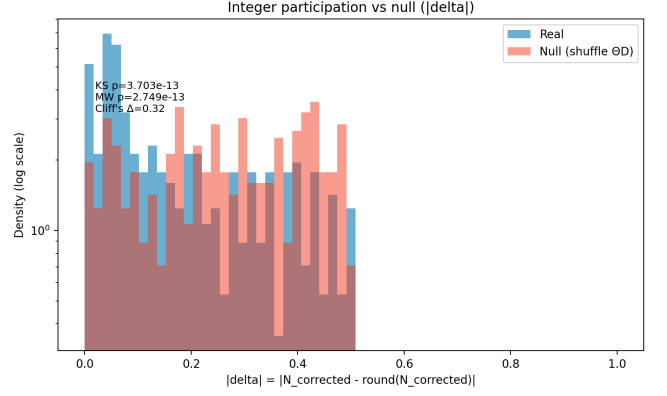


FIG. 2. Evidence of quantization. The distribution of locking residuals $|\delta| = |N_{\text{corr}} - \text{round}(N_{\text{corr}})|$ for the real data (blue) shows a sharp excess at $|\delta| \approx 0$ that is absent in the shuffled- Θ_D null (orange). Under the null, the probability of obtaining an equal or stronger concentration at zero deviation is $p < 10^{-13}$ (Kolmogorov–Smirnov and Mann–Whitney tests), and Cliff’s $\Delta \simeq 0.32$ indicates a medium-to-large effect size.

points colored by family. Despite considerable scatter, there is a clear positive monotonic trend: the Spearman correlation is $\rho \simeq 0.34$ with $p \simeq 5.6 \times 10^{-4}$. Materials with higher structural noise tend to deviate more from integer participation.

The same relationship is visible when comparing groups. Fig. 5 shows the distribution of z_ξ for the “almost-integer” group (top 20% lowest $|\delta|$) and the rest of the dataset. The almost-integer materials have markedly lower noise, with Cliff’s $\Delta \simeq -0.47$, a large negative effect size.

D. Base-frequency hierarchy across families

The calibrated base frequencies $f_{\text{base}}^{(k)}$ form a simple, interpretable pattern. Fig. 6 shows the median $f_{\text{base}}^{(k)}$ per family, ordered by increasing value, together with the global median $f_{\text{base,med}} \simeq 4.157$.

Iron-based superconductors cluster near $f_{\text{base}} \simeq 1$, suggesting that, in these systems, the mother frequency often participates in the fundamental mode and that F_m^* itself already behaves as an integer participation number. Heavy-fermion and oxide materials sit near $f_{\text{base}} \simeq 2$, consistent with a first harmonic. Binary, molecular and high-pressure superconductors, as well as type-I materials, concentrate around $f_{\text{base}} \simeq 4$, the same integer that controls the superfluid–roton ratio in ^4He . The independently calibrated f_{base} for the Superfluid family is $\simeq 4.0$, so the global median effectively recovers the ^4He scale when pooling all families.

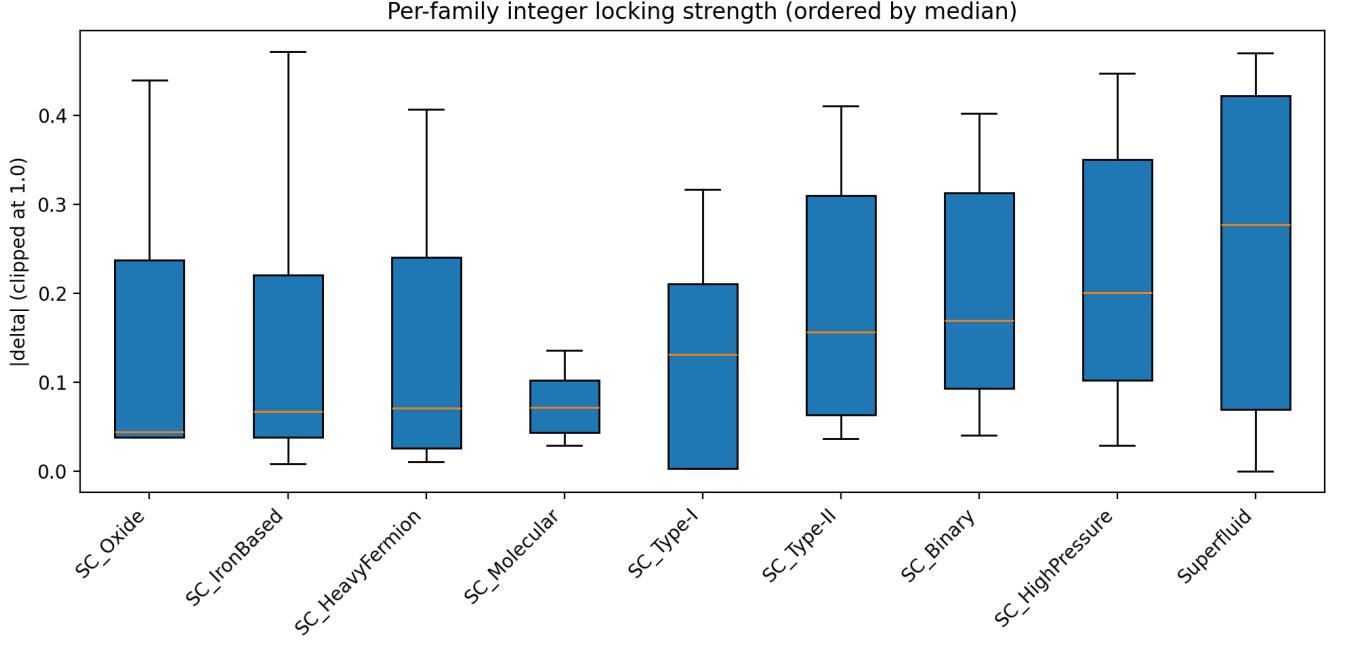


FIG. 3. Per-family distribution of locking residuals $|\delta|$ (clipped at 1.0). Boxes show interquartile ranges and medians; whiskers extend to 1.5 times the interquartile range. Families are ordered by increasing median residual.

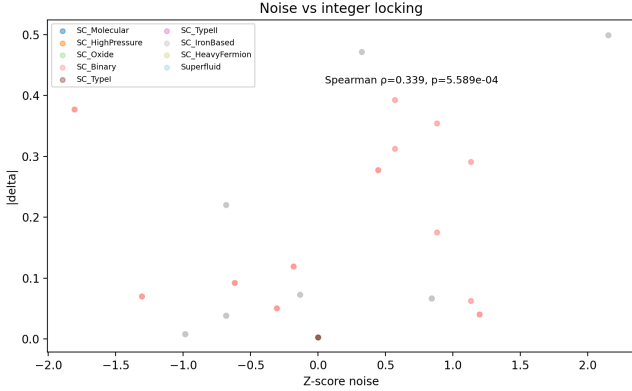


FIG. 4. Absolute locking residual $|\delta|$ vs structural-noise z_ξ . Colors indicate material families. The Spearman correlation is $\rho \simeq 0.34$ with $p \simeq 5.6 \times 10^{-4}$, indicating that higher structural noise is associated with weaker integer locking.

V. DISCUSSION

The analysis above establishes three empirical facts:

(i) After correcting for structural noise, the noise-corrected frequencies F_m^* admit a family-dependent base frequency $f_{\text{base}}^{(k)}$ such that the rescaled quantities N_{corr} cluster unusually close to integers.

(ii) The strength of this clustering, quantified by $|\delta|$, is modulated by structural noise both continuously (Fig. 4) and in group comparisons (Fig. 5), with low-noise mate-

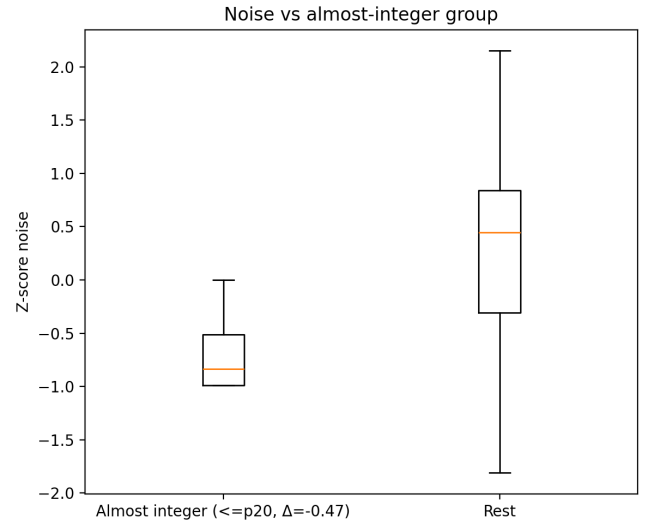


FIG. 5. Structural-noise z_ξ for materials in the “almost-integer” group (top 20% smallest $|\delta|$) and for the rest of the dataset. The almost-integer group is shifted toward lower noise, with Cliff’s $\Delta \simeq -0.47$.

rials displaying the most pronounced integer locking.

(iii) The calibrated base frequencies fall into a simple hierarchy of approximate harmonics: $f_{\text{base}} \sim 1$ for iron-based, ~ 2 for oxides and heavy fermions, and ~ 4 for binary, molecular, high-pressure and type-I superconductors, with a global median $\simeq 4.16$ very close to the $n = 4$ ratio characteristic of the roton branch in ^4He .

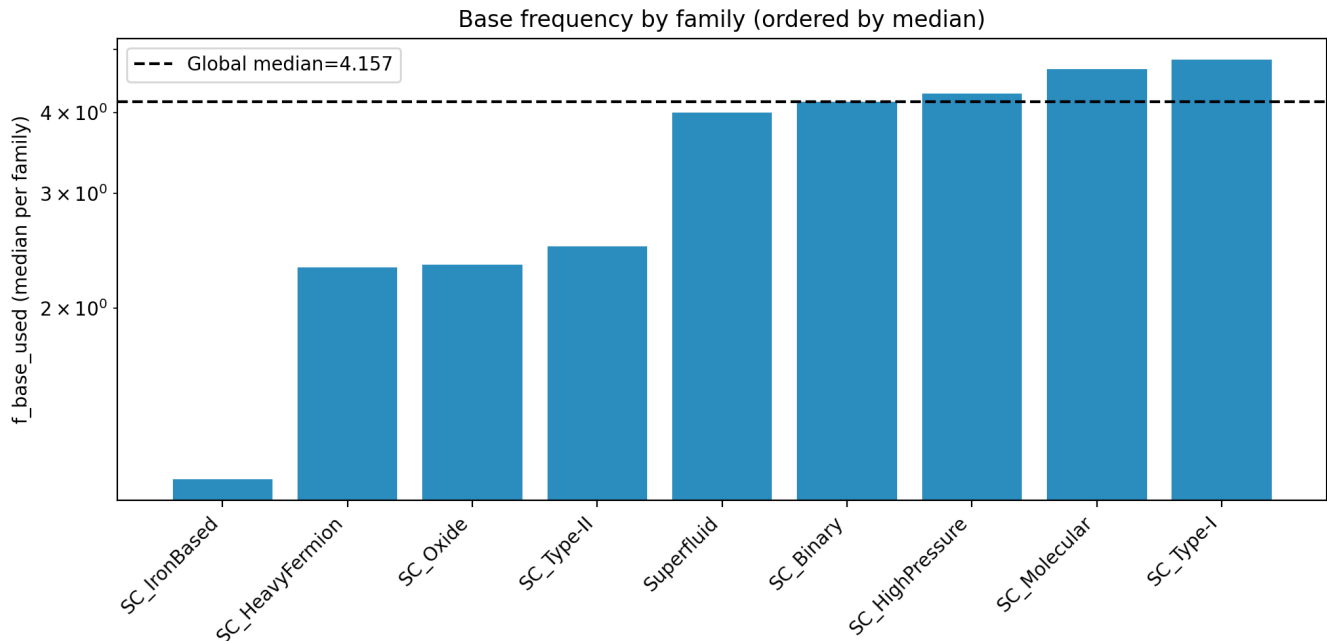


FIG. 6. Base frequency f_{base} per family (median over materials), ordered by increasing value. The dashed line marks the global median $f_{\text{base_med}} \simeq 4.157$. Families cluster around approximately harmonic values $f_{\text{base}} \sim 1, 2$ and 4 , with iron-based superconductors near the fundamental and binary / molecular / high-pressure materials near the $n = 4$ harmonic associated with ^4He .

One interpretation is that the mother frequency F_{m} is not an arbitrary thermal ratio but a coarse probe of an underlying oscillator network whose normal modes are discretized. Structural noise—arising from defects, frictional heating and pressure-induced distortions—broadens and shifts these modes. Where noise is low, the projection of F_{m}^* onto this discrete spectrum becomes visible as integer participation. Where noise is high, the same structure is smeared into the null.

We emphasize that our analysis is agnostic about the microscopic origin of these oscillators. They could arise from phase-coherent phonon modes, charge or spin density waves, or more exotic collective excitations. Our results do not prove a specific mechanism; they only constrain any viable mechanism to reproduce the observed, family-dependent integer structure once mapped into F_{m} .

At the same time, the numerical coincidence between the global f_{base} median and the ^4He roton ratio suggests that superconductivity and superfluidity may share a common “unit” of coherence, manifested at the level of a simple frequency ratio. Whether this reflects a deep universality class or a more mundane convergence of scales remains an open question.

VI. CONCLUSION

We have shown that a simple frequency ratio $F_{\text{m}} = \Theta_{\text{D}}/T_{\text{c}}$, when corrected for structural noise and rescaled

by a family-dependent base frequency, exhibits robust integer participation across a heterogeneous set of superconductors and liquid ^4He . The effect is statistically strong, structurally modulated and organized by approximate harmonics of a base frequency whose global scale matches that of ^4He .

Among all families, iron-based superconductors stand out as the clearest realization of this picture: with $f_{\text{base}} \simeq 1$, they operate in a natural resonance unit where the noise-corrected mother frequency F_{m}^* itself already acts as an integer participation number. In this sense, they behave as ideal, unscaled harmonic oscillators within the phenomenological framework developed here.

Rather than offering a full microscopic theory, our goal is to provide a compact phenomenological summary of patterns that any future theory ought to explain. The framework—mother frequency, structural noise, base-frequency calibration and integer participation—is deliberately minimal and can be applied to expanded datasets as they become available.

APPENDIX: STRUCTURAL-NOISE DIAGNOSTICS

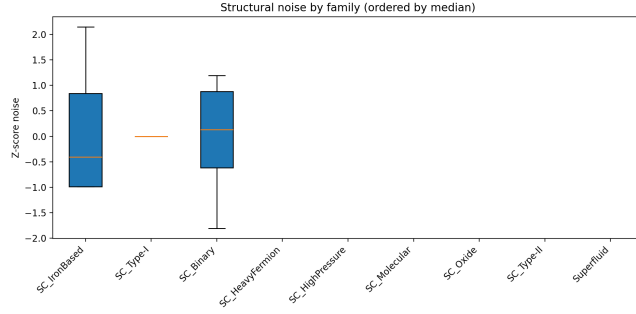


FIG. 7. Structural-noise z -score z_ξ by family (only families with a non-degenerate noise model are shown). Distributions are standardized within each family; boxes show interquartile ranges and medians.

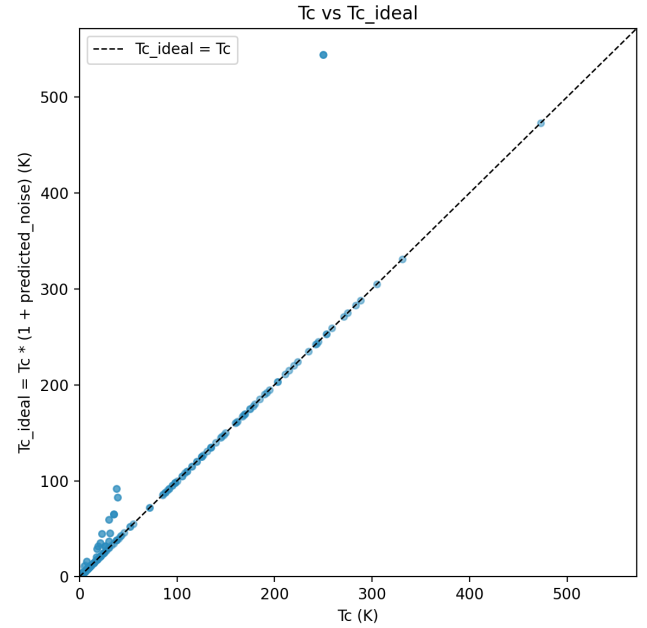


FIG. 8. Measured transition temperature T_c vs ideal temperature $T_c^{\text{ideal}} = T_c(1 + \hat{\xi})$. The dashed line indicates $T_c^{\text{ideal}} = T_c$. Most materials lie close to the identity, with a subset receiving larger upward corrections due to structural noise.

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